ABSTRACT

SupplyGuard

An AI-Powered Shipment Delay Prediction and Risk Analysis System

Global supply chain logistics faces severe operational disruptions and financial penalties caused by unforeseen delivery delays. Traditional monitoring relies on retrospective reporting and manual dispatch assessments, which fail to detect bottlenecks before shipments leave the warehouse facility. This project proposes SupplyGuard, an AI-powered system that applies machine learning to forecast late-delivery risk proactively from pre-dispatch order features.

The system uses the DataCo Smart Supply Chain Dataset (180,519 records) with 19 pre-dispatch features including shipping mode, scheduled transit days, geographic destinations, item pricing, and temporal attributes. After preprocessing, Random Forest and Support Vector Machine (SVM) classifiers are trained and evaluated on an 80:20 stratified split, with the tuned Random Forest model delivering 83.89% accuracy and 91.18% ROC-AUC.

The web application is built with Python Flask (backend) and responsive HTML5, CSS3, and JavaScript (frontend), featuring role-based access control for customers and logistics managers.

The expected outcome is an intelligent, user-friendly decision-support platform delivering instant shipment delay predictions and proactive risk mitigation guidance, improving logistics efficiency through the integration of machine learning and modern web technologies.

Dataset: https://data.mendeley.com/datasets/8gx2fvg2k6/5

Technology Stack :

Frontend: HTML5, CSS3, JavaScript, Bootstrap | Backend: Python, Flask

ML: Random Forest, Support Vector Machine (SVM), Scikit-learn, Pandas, NumPy

Tools: Visual Studio Code, Jupyter Notebook, Git, GitHub

References:

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[2] N. Rezki and M. Mansouri, "Integrating Machine Learning and Deep Learning for Enhanced Supplier Risk Prediction," Acta Logistica, vol. 11, no. 3, pp. 387–396, 2024.
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